

ARKANA



Odoo V18 Technical Training

Prasetiya Mulya University

arkana.co.id



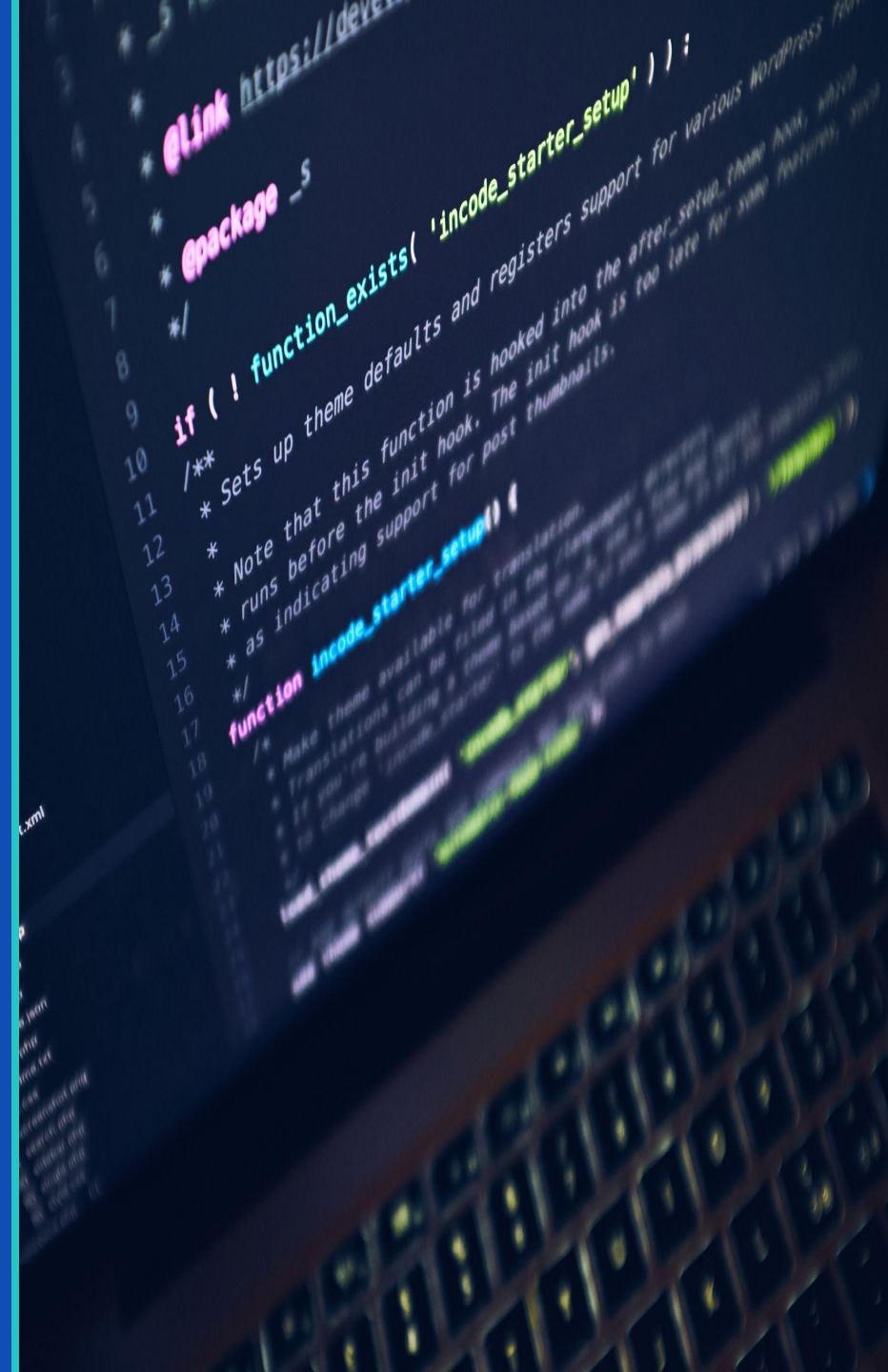
ARKANA



ODOO 18 · TECHNICAL FUNDAMENTAL

Day 4 — Reports & Integrations

QWeb PDF · Report Inheritance · Excel Export · REST Consumer ·
JSON-RPC / XML-RPC



Odoo Trainer Technical



Ardian Pramana

Lead Developer @Arkana

Rundown **Training** **Arkana**



- 08.30 - 10.00** : Sesi I
- 10.00 - 10.15** : Coffee Break
- 10.15 - 12.00** : Sesi II
- 12.00 - 13.15** : ISHOMA
- 13.15 - 15.20** : Sesi III
- 15.20 - 15.45** : Break Ashr
- 15.45 - 16.30** : Sesi IV

ARKANA

Follow our **Social Media**



@arkana_academy



@arkana_academy



Arkana Solusi Digital



@arkanasolusidigital

www.arkana.co.id

Day 3 Checkpoint

Day 3 — sudah jadi

- approval workflow berjenjang (L1/L2)
- security: groups, access rights, record rules
- monitoring views
- Reject Wizard + `_reject_with_reason()`

Day 4 — hari ini

- data dikeluarkan sebagai report (PDF, Excel)
- report di-extend tanpa menyalin ulang
- Odoo meng-consume REST API eksternal
- Odoo diakses via JSON-RPC / XML-RPC

Baseline = Day 3 final. State approval tetap utuh: draft · submitted · manager_approved · confirmed · done · rejected · cancelled.

Business Output vs System Integration

Business Output

- Output untuk manusia: dokumen & rekap.
- QWeb PDF — sertifikat / printout.
- Excel Export — data tabular untuk operasional.
- Part 1–3.

System Integration

- Komunikasi antar-sistem (mesin ke mesin).
- REST Consumer — Odoo memanggil API luar.
- JSON-RPC / XML-RPC — sistem lain mengakses Odoo.
- Part 4–5.



Dua sumbu Day 4: mengeluarkan output bisnis, dan menghubungkan Odoo dengan sistem lain.

Yang akan Anda kuasai

- A QWeb PDF Report** Certificate via `ir.actions.report` + QWeb (signature di-bundle).
- B Report Inheritance** Inherit report standar Odoo (Sales Order) via XPath — trainer demo.
- C Excel Export** Wizard TransientModel + `xlsxwriter` → `.xlsx` (`fields.Binary`).
- D REST API Consumer** Odoo memanggil REST API eksternal (timeout + error handling).
- E External API** JSON-RPC (Postman) vs XML-RPC (Python) — akses Odoo dari luar.

Alur lab hari ini



Tiap checkpoint = module installable yang sudah membawa seluruh approval workflow Day 3.

ARKANA

PART 1

QWeb PDF Report

Dokumen formal dari data Academy



Report rendering pipeline



Model menyediakan data → action menentukan jenis report → template QWeb menyusun layout → Odoo render ke PDF.

Anatomy of ir.actions.report

```
<record id="action_report_enrollment_certificate"
        model="ir.actions.report">
  <field name="model">academy.enrollment</field>
  <field name="report_type">qweb-pdf</field>
  <field name="report_name">
    academy_management.report_enrollment_certificate</field>
  <field name="binding_model_id"
        ref="model_academy_enrollment"/>
</record>
```

Poin penting

- report_type = qweb-pdf
- report_name = id template
- binding_model_id → tombol Print
- model = academy.enrollment

QWeb essentials: t-esc · t-if · t-foreach

t-esc

```
<span t-esc="value"/>
```

Cetak nilai / ekspresi ke output.

t-if

```
<p t-if="cond">...</p>
```

Render kondisional sebuah blok.

t-foreach

```
t-foreach="docs"
```

Ulangi blok untuk tiap record.

Certificate Report

Isi sertifikat

- Student · Course · Batch
- Enrollment Date
- State (approval state Day 3)
- Generated: `time.strftime('%Y-%m-%d')`

Signature di-bundle

- Blok signature langsung di report utama (bukan inheritance).
- 'Generated by Arkana Academy' + Trainer Signature.
- Certificate Reference = `doc.id`.
- Meaningful untuk enrollment confirmed.



Tanggal pakai `time.strftime('%Y-%m-%d')` — BUKAN `datetime.date.today()`.

QWeb PDF Report

1. Buat reports/academy_enrollment_certificate_report.xml (ir.actions.report + QWeb + signature).
2. Daftarkan di __manifest__.py (sebelum academy_menus.xml).
3. Upgrade, buka enrollment confirmed → Print → Enrollment Certificate.



SELESAI BILA

- PDF ter-generate
- data & signature benar
- tanggal pakai time.strftime



20–25 menit · full code di Lab Pack — Checkpoint A.

Report Inheritance

Extend report, jangan copy ulang



Why inherit, not copy

✗ Copy seluruh report

- Dua sumber kebenaran yang gampang divergen.
- Perbaikan upstream tak ikut.
- Berat dirawat saat report standar berubah.

✓ Inherit via XPath

- Hanya menambah blok yang dibutuhkan.
- Tetap mewarisi perubahan template dasar.
- Lebih kecil, fokus, mudah dirawat.

Standard Odoo report — Sales Order

```
<template id="report_saleorder_document_academy_note"
  inherit_id="sale.report_saleorder_document">
  <xpath expr="//div[class='page']"
    position="inside">
    <div class="mt-4">
      <p><strong>Processed by Arkana
Academy</strong></p>
    </div>
  </xpath>
</template>
```

Trainer demo / optional

- Butuh module sale ter-install.
- Inherit
sale.report_saleorder_document.
- Bukan blocker checkpoint lain.

XPath strategy & upgrade risk

Anchor stabil

- Pakai hasclass('page') / id yang stabil.
- Sisipkan position="inside" pada container.
- Uji ulang setelah upgrade Odoo.

Hindari yang rapuh

- Index node absolut (mudah patah).
- Bergantung struktur dalam yang sering berubah.
- XPath panjang menembus banyak level.



Report standar Odoo lebih kompleks — pilih XPath stabil agar tidak patah saat versi berubah.

Report Inheritance (Sales Order)

1. Buat reports/sale_order_report_inherit.xml (inherit_id sale.report_saleorder_document).
2. Pastikan module sale ter-install; tambahkan 'sale' ke depends + file ke data.
3. Upgrade, buka Sales → Sales Order → Print, verifikasi note Arkana.



SELESAI BILA

- note muncul di printout SO
- XPath stabil (hasclass page)
- tanpa menyalin report



15–20 menit · trainer demo / optional · Checkpoint B.

Excel Export

Data terstruktur untuk operasional



Excel export architecture



Wizard mengumpulkan filter → query data → tulis workbook in-memory → simpan sebagai Binary → user download.

Binary / base64 / download pattern

```
buffer = io.BytesIO()
wb = xlsxwriter.Workbook(buffer, {"in_memory":
True})
# ... tulis header + baris enrollment ...
wb.close()
self.file_data =
base64.b64encode(buffer.getvalue())
self.file_name = "enrollments.xlsx"
```

Kunci

- close() sebelum baca buffer
- base64 untuk fields.Binary
- filename → widget download

Manual xlsxwriter vs framework/addon

Manual xlsxwriter (lab)

- Memahami mekanisme export file di Odoo.
- Kontrol penuh atas isi & format.
- Cocok untuk export spesifik / sekali pakai.

Framework/addon (awareness)

- report_xlsx / addon matang → lebih reusable.
- Hemat waktu untuk banyak Excel report berulang.
- Pertimbangkan saat skala report bertambah.



Tidak ada satu cara paling benar — pilih sesuai kebutuhan & skala report.

Excel Export

1. Buat wizards/enrollment_export_wizard.py (TransientModel + xlsxwriter) & view-nya.
2. Tambah akses csv, menu Export Enrollments, import di wizards/__init__.py.
3. Daftarkan view + external_dependencies xlsxwriter; Export → download .xlsx.



SELESAI BILA

- enrollments.xlsx ter-download
- kolom benar termasuk State
- filter date/batch bekerja



25–30 menit · full code di Lab Pack — Checkpoint C.

PART 4

REST API Consumer

Odoo memanggil sistem lain



Consumer vs Provider

Consumer (Day 4)

- Odoo memanggil REST API eksternal.
- GET data course dari mock API.
- Validate payload + idempotent upsert.

Provider (di luar scope)

- Membuat endpoint REST sendiri di Odoo.
- `http.Controller` / `@http.route` — TIDAK dibahas.
- REST Provider bukan bagian Day 4.

Error handling & idempotent upsert

```
try:
    resp = requests.get(self.api_url, timeout=10)
    resp.raise_for_status()
except requests.exceptions.RequestException as error:
    raise UserError("Failed to call API: %s" % error)
existing = Course.search([("code", "=", code)], limit=1)
existing.write(vals) if existing else Course.create(vals) # upsert
```



timeout wajib · non-200 → UserError · upsert by code mencegah duplikat. Manager-only, tanpa sudo().

REST API Consumer

1. Buat wizards/import_external_courses_wizard.py (requests + timeout + UserError) & view.
2. Akses csv import HANYA Manager; menu dengan groups manager.
3. external_dependencies requests; jalankan mock server lalu Import.



SELESAI BILA

- course ter-import/upsert
- Manager-only (user biasa ditolak)
- tanpa sudo()



25–30 menit · full code di Lab Pack — Checkpoint D.

PART 5

Odoo External API: JSON-RPC vs XML-RPC

Sistem lain mengakses Odoo



JSON-RPC with Postman

```
POST /web/dataset/call_kw/academy.course/search_read
{
  "jsonrpc": "2.0", "method": "call",
  "params": {
    "model": "academy.course",
    "method": "search_read",
    "args": [[], ["name", "code", "level"]],
    "kwargs": {"limit": 10}
  }
}
```

3 langkah Postman

- POST /web/session/authenticate
- POST .../call_kw/.../search_read
- POST .../call_kw/.../create
- Cookie jar → session_id terbawa

JSON-RPC is not REST

REST

- Resource-oriented.
- Endpoint = resource.
- HTTP verbs: GET/POST/PUT/DELETE.
- Contoh: GET /api/courses/1

JSON-RPC

- Method-oriented.
- Payload menentukan method.
- model + method di body (call_kw).
- Contoh: academy.course · search_read



JSON-RPC bukan REST API — walau sama-sama HTTP & JSON, ia remote procedure call berbasis JSON.

XML-RPC with Python

```
common = ServerProxy("%s/xmlrpc/2/common" % URL)
uid = common.authenticate(DB, USERNAME, PASSWORD, {})
models = ServerProxy("%s/xmlrpc/2/object" % URL)
courses = models.execute_kw(DB, uid, PASSWORD,
    "academy.course", "search_read",
    [[[]], {"fields": ["name", "code", "level"], "limit": 10})
```



XML-RPC: authenticate → execute_kw(model, method, args). Script Python standar — cocok untuk integrasi server-to-server.

Same permission boundary

RPC tunduk pada keamanan yang sama

- JSON-RPC & XML-RPC memakai access rights & record rules yang sama dengan UI.
- Tidak ada jalan pintas: user RPC tetap dibatasi group & rule Day 3.



Gunakan integration user non-admin dengan hak minimum — jangan pakai admin untuk integrasi rutin.

External API: JSON-RPC vs XML-RPC

1. JSON-RPC (Postman): authenticate → search_read → create academy.course (payload di handout).
2. XML-RPC (Python): jalankan scripts/xmlrpc_search_read_courses.py & xmlrpc_create_course.py.
3. Uji dengan integration user non-admin; amati permission boundary.



SELESAI BILA

- JSON-RPC call_kw berhasil
- XML-RPC list/create berhasil
- akses tunduk pada group/rule

Kesalahan yang sering terjadi

Report / QWeb

- action salah model / report_name#template id
- QWeb variable tidak tersedia
- XPath terlalu fragile

Excel Export

- pakai sudo() tanpa alasan
- export semua data tanpa limit/domain

REST Consumer

- tanpa timeout / tak handle non-200
- duplicate karena tak ada external key
- wizard dibuka untuk user tanpa hak

External API

- menyebut JSON-RPC sebagai REST
- salah endpoint JSON-RPC
- XML-RPC gagal karena permission user

Production checklist

Report & Export

- report diuji dengan data kosong/edge
- access rights diuji dgn user non-admin
- export diuji dengan jumlah baris realistis
- tanpa sudo() yang tidak perlu

Integration

- API timeout & error handling diuji
- duplicate import dicegah (external key)
- RPC diuji dgn integration user non-admin
- endpoint & payload terdokumentasi

Academy kini dapat menghasilkan report & terintegrasi

A

QWeb PDF

Certificate + signature.

B

Inheritance

Inherit SO report
(demo).

C

Excel Export

TransientModel +
xlsxwriter.

D

REST Consumer

requests,
Manager-only.

E

External API

JSON-RPC &
XML-RPC.

Q & A

Terima kasih — sampai jumpa di Day 5.

Terima Kasih

Prasetiya Mulya University · Odoo V18 Technical Training

ARKANA

www.arkana.co.id